

Strategy–Action Synchronization AI (ISPS)

Intelligent Strategic Plan Synchronization System

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Background & Motivation

Background

- Organizations typically maintain strategic plans and action plans as separate documents.
- Strategic plans describe long-term goals and organizational vision, while action plans define short-term operational tasks.
- Because these documents are created independently, it becomes difficult to verify whether daily activities truly support strategic objectives.

Motivation

- Alignment between strategy and execution is usually checked manually through meetings and reviews, which are time-consuming, subjective, and not scalable.
- Traditional keyword-based analysis fails to capture semantic meaning, as the same intent can be expressed using different wording.
- Therefore, there is a strong need for an AI-based, objective, and scalable system that can automatically measure alignment, identify gaps, and support decision-making.

Problem Statement

- Despite the importance of strategic planning, organizations lack an objective and automated mechanism to evaluate whether their action plans effectively support strategic goals. Existing alignment checks rely heavily on manual review and human judgment, which introduces subjectivity and inconsistency.
- Furthermore, traditional text analysis approaches depend on keyword matching, which fails to capture semantic relationships between strategies and actions written using different terminology. As a result, organizations cannot quantitatively measure alignment, identify weak or missing actions, or continuously monitor strategic execution.
- This project addresses the problem of how to automatically, accurately, and scalably measure strategy–action alignment using semantic understanding and AI-driven techniques.

Project Objectives

- The primary objective of this project is to automatically evaluate the alignment between organizational strategic plans and action plans using semantic analysis techniques. Instead of relying on manual review, the system aims to provide an objective and measurable assessment of how well operational actions support strategic goals.
- In addition, the system is designed to identify weakly aligned or missing action items that fail to adequately support specific strategic objectives. By detecting these gaps, the system enables organizations to recognize areas where strategic execution needs improvement.
- Another key objective is to generate AI-driven improvement suggestions using Retrieval-Augmented Generation. These recommendations are grounded in existing action-plan context and help decision-makers refine current actions or introduce new ones.
- Finally, the project aims to visualize alignment results through an interactive dashboard, allowing both technical and non-technical stakeholders to easily understand alignment levels, coverage gaps, and recommendations.

Technologies Used

- Natural Language Processing (NLP)

Used to clean, normalize, and preprocess strategic and action plan text for accurate analysis.

- Sentence Embeddings

Converts text into semantic vectors that capture meaning beyond keywords.

- Vector Database (ChromaDB)

Stores embeddings and enables fast similarity search using cosine similarity.

- Retrieval-Augmented Generation (RAG)

Combines retrieved relevant actions with LLM reasoning to generate grounded recommendations.

- Large Language Model (LLM)

Generates human-readable explanations and improvement suggestions.

- Streamlit Dashboard

Provides an interactive interface to visualize alignment results and recommendations.

High-Level System Architecture

Strategic Plan + Action Plan



Text Preprocessing



Sentence Embeddings



Vector Database



Top-K Retrieval



Alignment Scoring



RAG Recommendations



Dashboard

Methodology

- Document Input

Strategic Plan and Action Plan are uploaded in PDF or JSON format.

- Text Preprocessing

Text is cleaned, normalized, and split into individual strategy and action statements.

- Embedding Generation

Each statement is converted into a semantic vector using a sentence-embedding model.

- Vector Storage

All embeddings are stored in a vector database for efficient retrieval.

- Similarity Matching (Top-K)

For each strategy, the most relevant action items are retrieved using cosine similarity.

- Alignment & Gap Analysis

Alignment strength is classified and missing or weakly aligned strategies are identified.

- RAG Recommendations

Retrieved actions are passed to an LLM to generate improvement suggestions.

- Dashboard Visualization

Results are displayed through charts, tables, and recommendations in the dashboard.

Alignment Analysis (Top-K Retrieval)

- Strategy as Query

Each strategic objective is treated as a search query.

- Vector Similarity Search

The vector database searches for semantically similar action items.

- Top-K Action Retrieval

The K most relevant actions are retrieved for each strategy.

- Cosine Similarity Scoring

Similarity scores indicate how closely actions support the strategy.

- Alignment Classification

Actions are categorized as strongly, partially, or weakly aligned.

- Interpretability

Retrieved actions clearly show why a strategy is considered aligned or not.

Results

- Strong Alignment Identified

Several strategic objectives were supported by highly relevant action items.

- Partial Alignment Detected

Some strategies had limited action support, indicating incomplete execution.

- Weak or Missing Alignment Found

Certain strategic goals had no strong supporting actions and were flagged as gaps.

- Quantitative Similarity Scores

Alignment was measured using cosine similarity, providing objective results.

- Clear Strategy–Action Mapping

The system clearly showed which actions support each strategy.

Evaluation

- Similarity Metric Used

Cosine similarity was used to measure semantic alignment between strategies and actions.

- Threshold-Based Classification

Similarity scores were grouped into strong, partial, and weak alignment categories.

- Manual Validation

Selected strategy–action pairs were reviewed to confirm meaningful alignment.

- Coverage Analysis

Evaluation measured how many strategies were supported by at least one strong action.

- RAG Output Assessment

Generated recommendations were checked for relevance and contextual grounding.

Dashboard Demonstration

- Alignment Overview Charts

Visualizes overall alignment levels across strategic objectives.

- Strategy–Action Mapping Table

Displays Top-K matched actions for each strategy with similarity scores.

- Coverage Gap Indicators

Highlights strategic objectives with weak or missing action support.

- AI-Generated Recommendations

Shows improvement suggestions produced using RAG.

- User-Friendly Interface

Enables non-technical users to explore results easily.

Security & Deployment

- Secure Document Handling

Strategic and action documents are processed in a controlled environment to avoid data leakage.

- Limited Data Exposure to LLMs

Only Top-K retrieved text is sent to the language model using RAG, not full documents.

- API Key Protection

Sensitive credentials are stored using environment variables, not hard-coded.

- Server-Side Processing

Similarity calculations and vector searches are performed securely on the server.

- Cloud-Ready Deployment

The system can be deployed using Streamlit Cloud or Hugging Face Spaces with HTTPS access.

Conclusion

- Objective Achieved

The system successfully evaluates alignment between strategic plans and action plans automatically.

- Effective Use of AI

Semantic embeddings and RAG enable meaningful alignment beyond keyword matching.

- Actionable Insights

The system identifies gaps and provides AI-generated improvement suggestions.

- Improved Transparency

Results are measurable, explainable, and easy to understand through the dashboard.

- Practical Applicability

The approach is suitable for real-world organizational decision-making.

Future Work

- Ontology / Knowledge Graph Integration

Add structured domain knowledge for deeper reasoning beyond similarity.

- Performance Metrics (KPIs)

Link strategy–action alignment with measurable organizational outcomes.

- Multilingual Support

Extend the system to analyze documents in multiple languages.

- Adaptive Threshold Learning

Automatically adjust alignment thresholds based on feedback and data.

- Enterprise Deployment

Add authentication, role-based access, and audit logging for large organizations.

Q & A

Thank You
Any Questions?